

Retail Sales & Operational Performance Analysis for Superstore

Project Overview

Project Title

Retail Sales & Operational Performance Analysis for Superstore

Domain

Retail / Business Intelligence / Supply Chain Analytics

Skills Demonstrated

- Exploratory Data Analysis (EDA)
 - Data Cleaning & Preprocessing
 - Business Intelligence & Visualization
 - Sales & Profitability Analysis
 - Customer Segmentation
 - Operational Optimization
 - Data-Driven Decision Making
 - SQL Query Writing
 - Power BI Dashboard Development
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1. Executive Summary

The objective of this project is to analyze Superstore transactional data to uncover trends, identify business performance gaps, and provide actionable recommendations to improve profitability and operational efficiency.

The project involved:

- Cleaning and preprocessing raw transactional data
- Performing SQL-based exploratory data analysis
- Building interactive Power BI dashboards
- Identifying profitable and unprofitable business areas
- Evaluating operational metrics such as shipping delay and returns
- Understanding the impact of discounts on profit margins

The analysis revealed several important business insights:

- Consumer segment generated the highest sales and profit.
- Copiers were the most profitable sub-category.
- Tables were the least profitable sub-category with negative profit.
- Office Supplies had the highest return rate among categories.
- High discounts frequently resulted in negative profit margins.
- Central region showed the highest average shipping delay.

The Power BI dashboard provides an interactive view of sales trends, profitability, regional performance, customer segmentation, and return analysis.

2. Problem Statement

The primary goal of this project is to analyze Superstore sales and operational data to:

- Identify profitable and loss-making products/categories
 - Evaluate regional sales performance
 - Understand customer segment behavior
 - Detect operational inefficiencies such as shipping delays
 - Measure the impact of discounts on profitability
 - Improve business decision-making using analytics
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3. Business Objectives

Key Objectives

1. Analyze sales and profitability trends.
 2. Identify high-performing and low-performing products.
 3. Evaluate return patterns and operational delays.
 4. Understand customer segment contribution.
 5. Build an interactive dashboard for business stakeholders.
 6. Provide strategic recommendations for business improvement.
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4. Stakeholders

The project insights are useful for:

- Sales Managers
- Regional Managers
- Supply Chain Teams

- Marketing Teams
 - Finance Teams
 - Business Executives
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5. Dataset Description

Key Fields Used

Field Name	Description
Order ID	Unique identifier for orders
Order Date	Date order was placed
Ship Date	Date order was shipped
Customer Name	Name of customer
Segment	Consumer / Corporate / Home Office
Region	East / West / South / Central
Category	Product category
Sub-Category	Product sub-category
Product Name	Product description
Sales	Revenue generated
Profit	Net profit
Discount	Discount percentage
Shipping Cost	Shipping expense
Return Status	Indicates returned orders
Shipping Delay	Calculated delivery delay
Profit Margin	Profit divided by sales

6. Data Cleaning & Preprocessing

Steps Performed

1. Removed Duplicate Records

- Checked duplicate Order IDs
- Removed duplicate transactional entries

2. Corrected Data Types

- Converted Order Date and Ship Date into proper date format
- Converted Sales, Profit, Shipping Cost into numeric format

3. Handled Missing Values

- Checked blanks in Profit, Shipping Cost, and Return Status
- Cleaned inconsistent or missing entries

4. Standardized Categorical Values

- Standardized:
- Region labels
- Category labels
- Segment names

5. Created Calculated Columns

Shipping Delay

Shipping Delay = Ship Date - Order Date

Profit Margin

Profit Margin = Profit / Sales

7. SQL-Based Exploratory Data Analysis (EDA)

Task 1: Top 10 Products by Total Sales

SQL Query

```
select product_name,  
       sum(sales) as total_sales
```

```
from products1
group by product_name
order by total_sales desc
limit 10;
```

Key Findings

- Canon imageCLASS 2200 Advanced Copier generated the highest sales.
- High-ticket technology and office equipment products dominated top sales rankings.
- Premium office products contributed significantly to revenue.

Task 2: Average Shipping Delay by Region

SQL Query

```
select region,
       round(avg(shipping_delay),2) as average_shipping_delay
from products1
group by region;
```

Results

Region	Average Shipping Delay
South	3.96
West	3.95
East	3.92
Central	4.04

Key Findings

- Central region had the highest shipping delay.
- East region had the lowest average shipping delay.
- Shipping operations in Central region may require optimization.

Task 3: Returned Orders with Shipping Delay Greater than 5 Days

SQL Query

```
select *  
from products1  
where shipping_delay > 5  
and return_status = 'Yes';
```

Key Findings

- Delayed shipments were strongly associated with returned orders.
 - Standard Class shipping mode appeared frequently in delayed returns.
 - Operational inefficiencies may be impacting customer satisfaction.
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Task 4: Monthly Total Sales and Profit

SQL Query

```
select extract(month from order_date) as month,  
       sum(sales) as total_sales,  
       sum(profit) as total_profit  
from products1  
group by month  
order by month;
```

Key Findings

- November and December recorded the highest sales.
 - Peak profitability occurred during the year-end season.
 - January and February showed lower business performance.
 - Strong seasonal sales trend identified.
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Task 5: Total Profit and Sales by Customer Segment

SQL Query

```
select segment,  
       sum(profit) as total_profit,  
       sum(sales) as total_sales
```

```
from products1
group by segment;
```

Results

Segment	Total Profit	Total Sales
Consumer	63164	491873.83
Corporate	39377	303219.21
Home Office	29902	197125.13

Key Findings

- Consumer segment generated the highest sales and profit.
- Home Office segment contributed the least revenue.
- Consumer customers are the primary revenue drivers.

Task 6: Most and Least Profitable Sub-Categories

Least Profitable Query

```
select sub_category,
       sum(profit) as least_profit
from products1
group by sub_category
order by least_profit asc
limit 1;
```

Most Profitable Query

```
select sub_category,
       sum(profit) as most_profit
from products1
group by sub_category
order by most_profit desc
limit 1;
```

Results

Type	Sub-Category	Profit
Most Profitable	Copiers	24563

Type	Sub-Category	Profit
Least Profitable	Tables	-9084

Key Findings

- Copiers delivered the highest profitability.
- Tables generated significant losses.
- Furniture category requires pricing and discount review.

Task 7: Return Rate by Category

SQL Query

```
select category,
       round((count(case when return_status='Yes' then 1 end)
              *100.0/count(*)),2) || '%' as return_rate_percentage
from products1
group by category;
```

Results

Category	Return Rate
Furniture	5.03%
Office Supplies	5.33%
Technology	3.80%

Key Findings

- Office Supplies had the highest return rate.
- Technology products had the lowest return rate.
- Return management needs attention in Office Supplies.

Task 8: Orders with Negative Profit and High Discount

SQL Query

```
select *
from products1
```

```
where profit < 0
and discount > 0.30;
```

Key Findings

- Heavy discounts often resulted in negative profits.
- Many unprofitable orders belonged to Office Supplies.
- Excessive discounting reduced overall profitability.

Task 9: Sales and Returns by Discount

SQL Query

```
select sum(sales) as total_sales,
       count(case when return_status='Yes' then 1 end)
       as total_number_of_returns,
       discount
from products1
group by discount;
```

Key Findings

- Higher discounts showed increased return counts.
- Discount level 0.20 generated substantial sales volume.
- Very high discount levels correlated with lower profitability.

Task 10: Average Sales of Returned vs Non-Returned Orders

SQL Query

```
select return_status,
       round(avg(sales),2)
from products1
group by return_status;
```

Results

Return Status	Average Sales
No	216.24
Yes	211.11

Key Findings

- Returned and non-returned orders had similar average sales.
 - Return behavior may depend more on operations than sales value.
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Task 11: States with Sales Over \$500,000

SQL Query

```
select state,
       sum(sales) as total_sales,
       sum(profit) as total_profit,
       count(*) as number_of_orders
from products1
group by state
having sum(sales) > 500000;
```

Observation

- No states crossed the \$500,000 sales threshold.

Insight

- Sales distribution is diversified across multiple states.
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Task 12: Customers with More Than 5 Orders

SQL Query

```
select count(*)
from (
  select customer_name
  from products1
  group by customer_name
  having count(*) > 5
) as customers;
```

Result

- 411 customers placed more than 5 orders.

Key Findings

- Strong repeat customer base exists.
 - Customer retention opportunities are available.
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Task 13: Customer-Wise Total Sales and Orders

SQL Query

```
select customer_name,
       sum(sales) as total_sales,
       count(order_id) as total_orders
from products1
group by customer_name;
```

Key Findings

- Some customers generated significantly higher order counts.
 - High-value repeat customers can be targeted for loyalty programs.
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Task 14: Unique Products Sold per Category

SQL Query

```
select category,
       count(distinct product_name) as number_product_name
from products1
group by category;
```

Results

Category	Unique Products
Furniture	357
Office Supplies	957
Technology	349

Key Findings

- Office Supplies had the largest product variety.
- Technology category had fewer but higher-value products.

Task 15: Average Discount by Region and Segment

SQL Query

```
select segment,
       region,
       avg(discount)
from products1
group by segment, region;
```

Key Findings

- Corporate segment in Central region received highest discounts.
- West region showed comparatively lower discount levels.
- Higher discounts may be affecting regional profitability.

8. Business-Focused SQL Data Extraction

Objective

Extract a clean and analysis-ready subset of business-critical data.

SQL Query

```
select *
from products1
where order_date >= '2021-01-01'
and sales > 200
and segment in ('Consumer', 'Corporate')
and region in ('East', 'South', 'Central')
and (
    return_status = 'Yes'
    or profit < 0
    or discount >= 0.20
)
order by order_date
limit 50000;
```

Purpose of Extraction

- Focus on high-value transactions

- Capture operational risk cases
 - Analyze discount impact
 - Identify loss-making orders
 - Build meaningful dashboard insights
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9. Power BI Dashboard Development

Dashboard Components

KPI Cards

- Total Sales
- Total Profit
- Average Shipping Delay
- Return Rate

Visualizations Used

Visualization	Purpose
Stacked Column Chart	Sales by Region and Category
Line Chart	Profit Trend Over Time
Scatter Plot	Discount vs Profit
Donut Chart	Return Rate by Sub-Category
Map Visualization	Sales by State
Slicers	Region, Segment, Date, Return Status

10. DAX Measures Used

Total Sales

```
Total Sales = SUM(products1[Sales])
```

Total Profit

```
Total Profit = SUM(products1[Profit])
```

Profit Margin

```
Profit Margin = DIVIDE(SUM(products1[Profit]), SUM(products1[Sales]))
```

Return Rate

```
Return Rate =  
DIVIDE(  
    COUNTROWS(FILTER(products1, products1[Return_Status] = "Yes")),  
    COUNTROWS(products1)  
)
```

Average Shipping Delay

```
Average Shipping Delay = AVERAGE(products1[Shipping_Delay])
```

11. Dashboard Insights

Sales Analysis

- East and West regions generated strong sales performance.
- Consumer segment contributed the highest revenue.
- Technology products delivered higher-value transactions.

Profitability Analysis

- Copiers and Accessories were highly profitable.
- Tables created major losses.
- High discount strategies negatively affected profit margins.

Operational Insights

- Central region experienced higher shipping delays.
- Returned orders were often associated with delayed shipping.
- Standard Class shipping showed more operational issues.

Customer Insights

- Large number of repeat customers identified.
- Consumer customers dominated order volume.

- High-value customers can be targeted for retention campaigns.
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12. Strategic Recommendations

1. Reduce High Discount Dependency

- Avoid excessive discounts above 30%.
- Implement controlled promotional strategies.
- Focus on value-based pricing.

2. Improve Shipping Operations

- Optimize logistics in Central region.
- Reduce shipping delays to minimize returns.
- Improve warehouse and delivery coordination.

3. Focus on Profitable Categories

- Increase investment in Copiers and Technology.
- Promote high-margin products.
- Reevaluate pricing for Tables and low-profit products.

4. Strengthen Customer Retention

- Launch loyalty programs for repeat customers.
- Provide personalized offers for Consumer segment.
- Use customer purchase history for targeted marketing.

5. Optimize Return Management

- Investigate reasons behind Office Supplies returns.
 - Improve product quality checks.
 - Enhance packaging and shipping reliability.
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13. Business Impact

This project helps the business:

- Improve operational efficiency
- Increase profitability
- Reduce unnecessary discounts
- Minimize returns and delays

- Enhance customer satisfaction
 - Support data-driven decision making
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14. Project Outcome

Final Deliverables

1. SQL-Based Business Analysis

- Multiple SQL queries for business insights
- Operational and profitability analysis

2. Power BI Dashboard

- Interactive business dashboard
- KPI tracking and visual storytelling

3. Business Report

- Structured analysis report
 - Strategic recommendations
 - Data-driven conclusions
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15. Conclusion

The Retail Sales & Operational Performance Analysis project successfully identified key business patterns across sales, profitability, customer behavior, returns, and operational efficiency.

The analysis demonstrated that:

- Consumer customers are the strongest revenue contributors.
- Excessive discounts significantly impact profitability.
- Shipping delays contribute to higher return rates.
- Some sub-categories require immediate strategic attention.

Using SQL and Power BI together enabled effective business intelligence reporting and helped transform raw transactional data into actionable insights.

The final dashboard and report provide management teams with a strong analytical foundation for improving business performance and operational decision-making.

16. Tools & Technologies Used

Tool	Purpose
PostgreSQL	SQL Analysis
Excel	Data Cleaning
Power BI	Dashboard & Visualization
SQL	Data Querying
DAX	KPI Measures

17. Future Enhancements

- Add predictive sales forecasting models
- Implement customer churn analysis
- Build automated reporting pipelines
- Integrate real-time sales monitoring
- Add machine learning-based recommendation systems

18. Appendix

SQL Concepts Used

- GROUP BY
- ORDER BY
- HAVING
- Aggregate Functions
- CASE WHEN
- COUNT DISTINCT
- Filtering
- Date Functions
- Subqueries

Business KPIs Tracked

- Total Sales
- Total Profit
- Profit Margin
- Return Rate
- Average Shipping Delay

- Sales by Segment
 - Regional Performance
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End of Report